Answer Key: Vision International School - Final Exam (Mathematics)

SECTION – A

Q1. Multiple Choice Questions (MCQs)

1. **(c)** 176°F
   - *Calculation:* $(80 \times \frac{9}{5}) + 32 = 144 + 32 = 176$
2. **(c)** 1000
3. **(c)** Acute
4. **(d)** Circle
5. **(d)** Circumference
6. **(d)** 1000
7. **(b)** 49 cm²
   - *Calculation:* $7 \times 7 = 49$
8. **(b)** 1:100
   - *Reasoning:* 1 m = 100 cm, so 1 cm to 100 cm is 1:100.
9. **(a)** Bars
10. **(b)** ×2
    - *Pattern:* Powers of 2 (2, 4, 8, 16, 32...)

Q2. Fill in the blanks 11. **5** kg 12. **3** (three) 13. **Radius** 14. **Length** 15. **Pictures** (or **Symbols/Images**) 16. **Equal**

Q3. Match the Following

1. Deca $\rightarrow$ **c. 10/1** (10)
2. Deci $\rightarrow$ **a. 1/10** (0.1)
3. Centi $\rightarrow$ **e. 1/100** (0.01)
4. Milli $\rightarrow$ **d. 1/1000** (0.001)
5. Hecto $\rightarrow$ **b. 100/1** (100)

Q4. True / False 16. **False** (A right angle is 90°, a straight angle is 180°) 17. **True** 18. **True** 19. **False** (Sum of digits is 44, which is not divisible by 9) 20. **True**

SECTION – B

Q5. Convert (a) **2.5 km** (2500 ÷ 1000) (b) **3500 mL** (3.5 × 1000) (c) **10 g** (0.01 × 1000)

Q6. Convert the temperature (a) **36.67°C** (approx) * *Calculation:* $(98 - 32) \times \frac{5}{9} = 66 \times \frac{5}{9} \approx 36.666...$ (b) **68°F** * *Calculation:* $(20 \times \frac{9}{5}) + 32 = 36 + 32 = 68$ (c) **2.22°C** (approx) * *Calculation:* $(36 - 32) \times \frac{5}{9} = 4 \times \frac{5}{9} \approx 2.22...$

Q7. Find (a) Radius = **1.25 cm** $(2.5 \div 2)$ (b) Diameter = **3.6 cm** (1.8×2) (c) Supplement of: * (a) 65° = **115°** $(180 - 65)$ * (b) 90° = **90°** $(180 - 90)$ (d) Complement of: * (a) 25° = **65°** ($90 - 25$) * (b) 55° = **35°** $(90 - 55)$

Q8. Types of triangles classified by side length

- One with all equal sides = **Equilateral triangle**
- One with two equal sides = **Isosceles triangle**
- One with no equal sides = **Scalene triangle**

SECTION – C

Q9. Convert the following (a) **5750 m** (5.75×1000) (b) **4 kg 680 g** (c) **2500 mL** (d) **3.5 m** ($350 \div 100$) (e) **4.2 L** $(4200 \div 1000)$

Q10. Find it

1. Area of rectangle (12 × 7) = **84 cm²**
2. Area of rectangle (25 × 18) = **450 cm²**
3. Area of square (15 × 15) = **225 cm²**
4. Volume (8 × 3 × 2) = **48 cm³**
5. Volume of cube $(5 \times 5 \times 5)$ = **125 cm³**
6. Perimeter of rectangle $(2 \times (10 + 5))$ = **30 cm**
7. Find $\angle C$: $180^\circ - (50^\circ + 60^\circ) = 180^\circ - 110^\circ$ = **70°**
8. Perimeter of triangle $(7 + 9 + 10)$ = **26 cm**

Q11. Solve it A] *Note: Requires visual image to estimate exactly, but generally:*

- Small angle (<90°) = Acute
- L-shape (90°) = Right
- Wide angle (>90°) = Obtuse

B]

- OA and OB are called **Arms** (or Sides) of the angle.

- Point 'O' is called **Vertex**.
- Name the angle: $\angle AOB$ (or $\angle BOA$)

C]

- a) $\angle P$ = **90** Degrees
- b) The measure of $\angle R$ is **90 Degrees**
- c) PQ is parallel to **SR** (or RS)
- d) PS is **Parallel** (or Equal) to QR

Q12. Pizza Chart Data *(Note: The specific numbers require reading the bar chart image, which was not visible in the text file. Below is the method to solve each.)*

1. Look at the height of the bar for **Thursday**.
2. Look for the **shortest** bar.
3. Look for the **tallest** bar.
4. Look for the bar that lines up with the number **25** on the axis.
5. Add the value of **Monday** + Value of **Wednesday**.
6. Compare the heights of Monday and Friday.
7. Subtract: (Value of **Thursday**) - (Value of **Friday**).
8. Add the values of **all 5 days** together.

SECTION – D (Long Answer)

Q12. Word Problems *(Note: Duplicate Q12 numbering in original paper)*

(a) Rope Length

- Total length: 5 km = 5000 m
- Cut: 1250 m
- Left: $5000 - 1250$ = **3750 m** (or **3.75 km**)

(b) Milk Volume

- Total milk: 12 L = 12000 mL
- Sold: 3500 mL
- Left: $12000 - 3500$ = **8500 mL** (or **8.5 L**)

(c) Volume of Room

- Formula: Length × Width × Height
- $10 \times 6.5 \times 5 = 65 \times 5$

- Volume = **325 m^3**

(d) Total Oil Quantity

- Tin 1: 10 L 150 mL
- Tin 2: 05 L 750 mL
- Tin 3: 03 L 875 mL
- *Sum of mL:* $150 + 750 + 875 = 1775$ mL
- *Convert mL:* 1775 mL = 1 L 775 mL
- *Sum of L:* $10 + 5 + 3 = 18$ L
- *Total:* $18\ \mathrm{L} + 1\ \mathrm{L}\ 775\ \mathrm{mL} =$ **19 L 775 mL**

(e) Square Numbers

- Next square number after 121 (11×11):
  - $12 \times 12 =$ **144**
- Next square number after 400 (20×20):
  - $21 \times 21 =$ **441**